

Title

Progressive resistance training of inspiratory muscles in non-dialysis chronic kidney disease: study protocol

Abstract

Background: Patients with pre-dialysis chronic kidney disease often develop thoracic volume restriction and respiratory muscle weakness, potentially contributing to exercise limitation and dyspnea. Whether progressive inspiratory muscle training can improve respiratory physiology in this population remains unclear. **Methods:** Thirty adults with CKD stages 3-5 on conservative treatment will be randomly assigned to two training protocols: experimental group (EG, n=15) receiving progressive resistance loading up to 50% of maximal inspiratory pressure, increased weekly, or control group (CG, n=15) performing fixed 5 cmH₂O loading without progression. Both groups train daily for 8 weeks. **Assessment timepoints:** baseline and week 8. **Primary outcome:** thoracic cavity volumes (pulmonary, rib cage, abdominal compartments) measured via non-invasive optoelectronic plethysmography. **Secondary outcomes:** maximal inspiratory and expiratory pressures measured via standard manometry. **Safety** will be monitored throughout including vital signs and adverse symptoms. **Inclusion:** confirmed CKD stage 3-5, stable renal function and medications ≥ 4 weeks, ability to perform breathing maneuvers. **Study design:** randomized controlled trial with parallel-group design.